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2025 Kriged Biomass Estimate and Sensitivities

NOAA Northwest Fisheries Science Center, Fisheries Engineering & Acoustic Technologies

DFO Pacific Region, Institute of Ocean Sciences and Pacific Biological Station



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Outline

- Biomass Estimation/Kriging
  - *2025 Biomass estimate*
  - *2025 Parameters for variogram + kriging*
    - *Sensitivity analysis*
- CV analysis
  - *Kriging*
  - *Jolly-Hampton*
- Results
  - *Extrapolation vs non-extrapolation*
  - *US & CAN*



2025 biomass estimates (backconverted to EK60)

Base case

Kolmogorov-Smirnov (K-S) stratification (post-survey)

*Geostatistics based estimate (Kriging) with **limited extrapolation** and **tapering** (recommended in 2015 by Survey & JTC)*

Extrapolation	Un-kriged Biomass (mmt)	Total kriged Biomass (mmt)	Relative Difference	CAN kriged Biomass (mmt)	Percentage of biomass in CAN	CV (ALL)	CV (US)	CV (CAN)
without	0.620	0.6406	-1.76%	0.005	0.70%	8.31%	6.76%	701.09%
with		0.6521	0.00%	0.005	0.77%	15.49%	12.82%	1282.27%

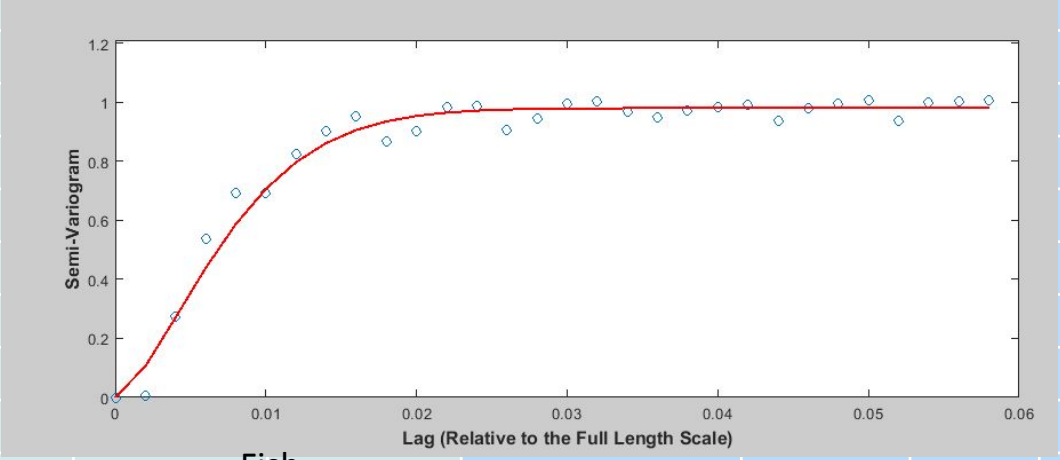
base case parameters

Semi-variogram & kriging parameters

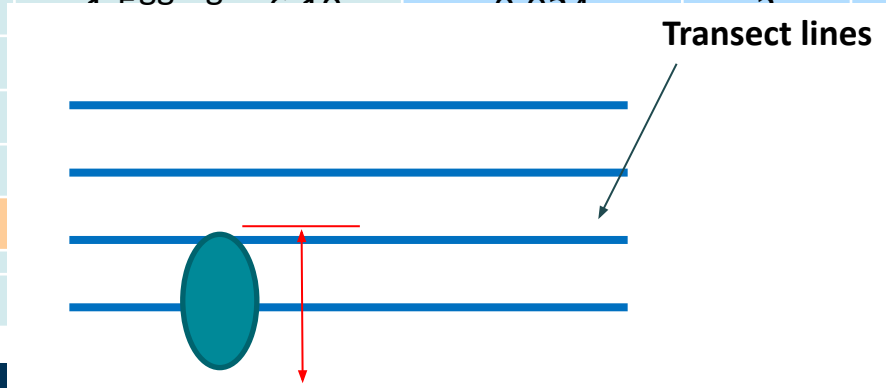
Semi-variogram						Kriging				
Year	nugget	sill	normalized length scale	power	aspect ratio	search radius	kmin	kmax	N-S length scale (nmi)	E-W length scale (nmi)
1995	0	0.96	0.006	1.5	6.2	0.018	3	10	7.23	1.16
1998	0	0.84	0.007	1.5	7.18	0.021	3	10	9.01	1.25
2001	0	0.94	0.006	1.5	6.15	0.018	3	10	6.54	1.06
2003	0	0.96	0.007	1	6.26	0.021	3	10	7.83	1.25
2005	0	0.89	0.007	1	6.38	0.021	3	10	7.98	1.25
2007	0	0.96	0.0065	1	6.39	0.0195	3	10	7.48	1.17
2009	0	0.9	0.008	1.3	6.33	0.024	3	10	9.12	1.44
2011	0	0.94	0.008	1.5	5.97	0.024	3	10	9.43	1.58
2012	0	0.84	0.008	1.5	6.00	0.024	3	10	9.37	1.56
2013	0	0.9	0.006	1.2	6.21	0.018	3	10	7.15	1.15
2015	0	0.98	0.006	2	6.16	0.018	3	10	8.08	1.31
2017	0	0.84	0.008	1.5	6.19	0.024	3	10	9.76	1.58
2019	0	0.91	0.007	1.5	6.06	0.021	3	10	8.57	1.41
2021	0	0.98	0.0085	1.5	6.42	0.026	3	10	10.33	1.46
2023	0	0.89	0.0055	1.75	6.24	0.0165	3	10	6.72	1.08
2025	0	0.97	0.0055	1.2	6.47	0.0165	3	10	6.68	1.03
Average	0	0.92	0.007	1.4	6.29	0.021	3	10	8.27	1.32

Semi-variogram & kriging parameters

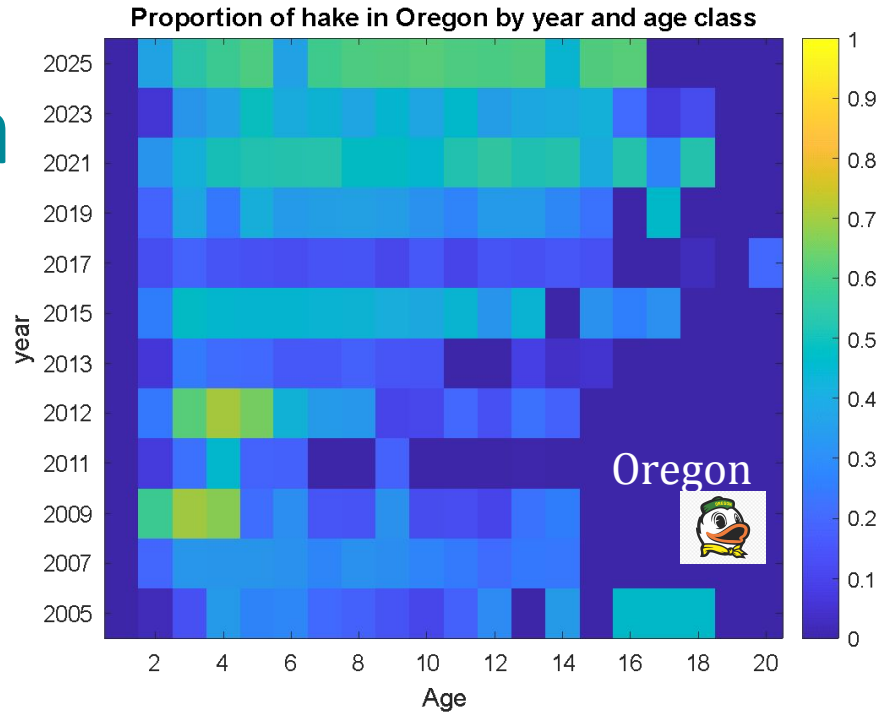
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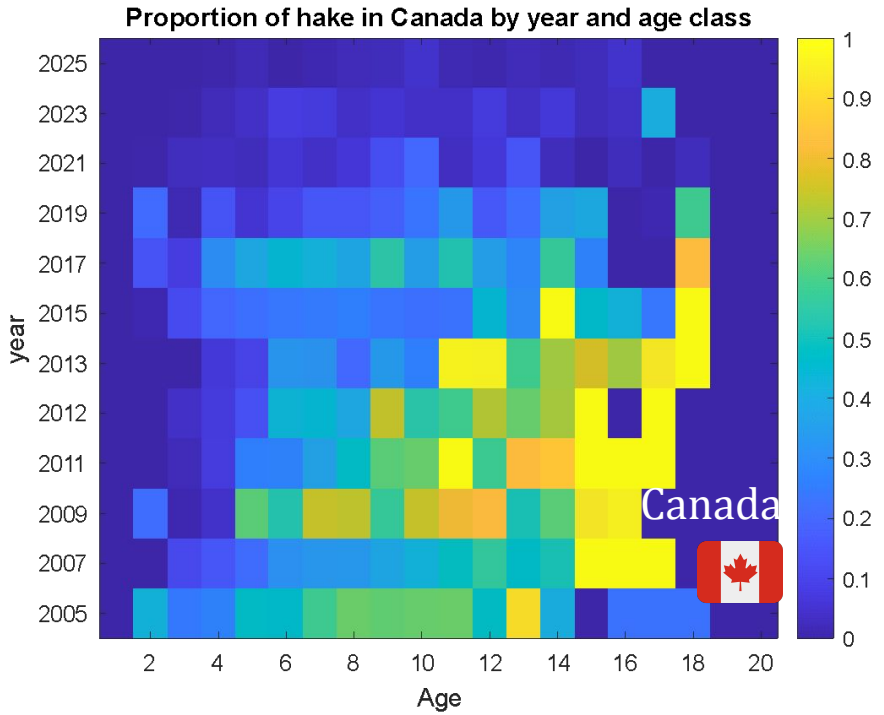


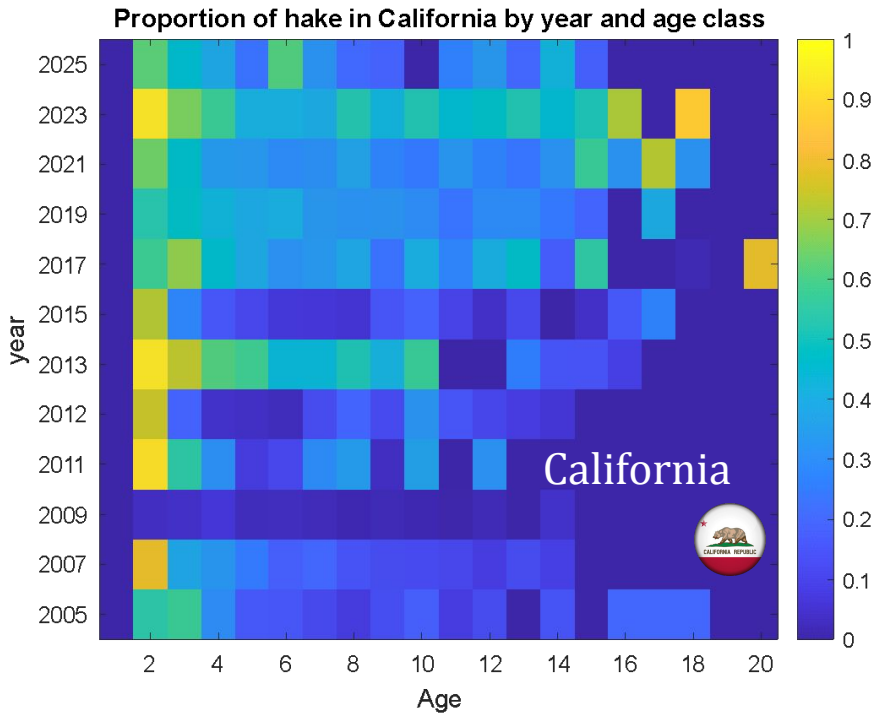
Fish aggregation

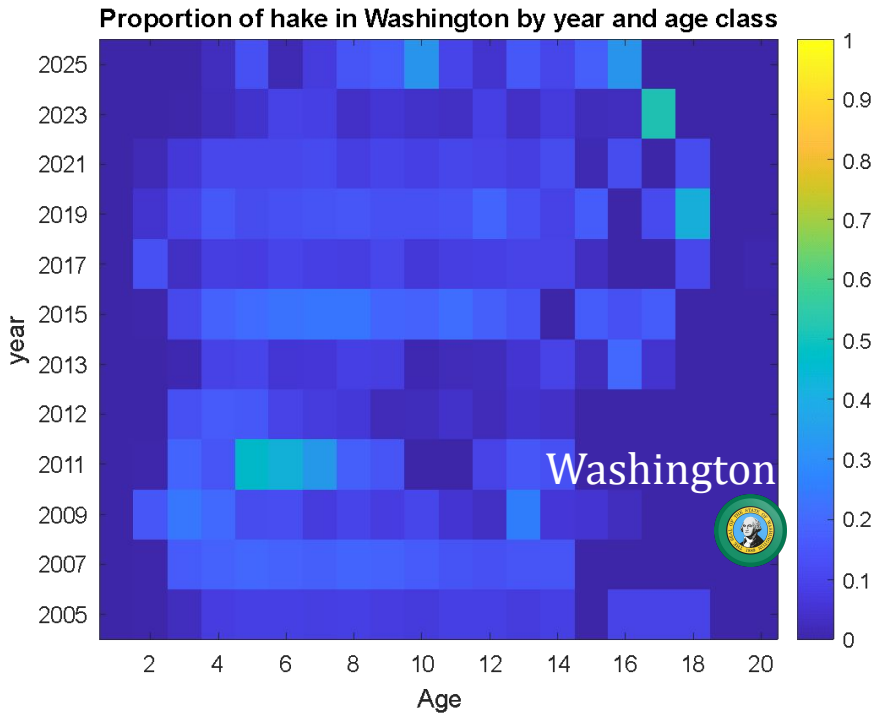


Age & Area


Proportion of hake in Oregon by year and age class
year
Age
Oregon


Proportion of hake in Canada by year and age class
year
Age
Canada


Proportion of hake in California by year and age class
year
Age
California


Proportion of hake in Washington by year and age class
year
Age
Washington

Sensitivity analyses

Semi-variogram parameters

- *Depend on the collected (un-kriged) NASC data*
- *Nugget, sill, length scale, power, and aspect ratio*

Kriging parameters

- *Search radius (SR)*
- *Minimum number of data points within SR (kmin)*
- *Maximum number of data points within SR (kmax)*



Photo by Rebecca Thomas

Semi-variogram parameters sensitivities (EK80)

	Parameters									
Semi-variogram	**chosen parameters**	**nugget**		**sill**		**length scale**		**power**		**aspect ratio**	
nugget	**0.00**	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
sill	**0.97**	0.97	0.95	0.99	0.97	0.97	0.97	0.97	0.97	0.97
length scale	**0.0055**	0.0055	0.0055	0.0055	0.0045	0.0065	0.0055	0.0055	0.0055	0.0055
power	**1.22**	1.22	1.22	1.22	1.22	1.22	0.92	1.52	1.22	1.22
aspect ratio	**6.47**	6.47	6.47	6.47	6.47	6.47	6.47	6.47	3.12	12.48
total biomass (mmt)	**0.615**	0.613	0.615	0.615	0.609	0.616	0.614	0.618	0.509	0.637
CV	**15.49%**	15.60%	15.33%	15.65%	15.92%	15.23%	15.40%	15.44%	20.25%	13.69%
relative difference	**0.00%**	**-0.37%**	**0.00%**	**0.00%**	**-1.00%**	**0.07%**	**-0.13%**	**0.47%**	**-17.49%**	**3.48%**

Most relative differences due to variation in variogram parameters are less than 4%; well below the assessment CV (~30%)

Exception: aspect ratio

Numbers in blue are those changing in the sensitivity

Kriging parameters

Search radius (SR) = 3 x length scale

Is dynamic and depends on the data structure

kmin = 3

Provides a reasonable variance estimate

kmax = 10

Provides stable variance estimate



Photo by John Pohl

Kriging parameters sensitivities (EK80)

	Final parameters	Search Radius		Kmin		kmax	
Search Radius	**0.0165**	**0.0083**	**0.0330**	0.0165	0.0165	0.0165	0.0165
kmin	**3**	3	3	**1**	**5**	3	3
kmax	**10**	10	10	10	10	**5**	**20**
Total Biomass (mmt)	**0.615**	0.497	0.631	0.616	0.614	0.624	0.623
CV	**15.49%**	19.61%	14.81%	16.24%	14.96%	15.47%	15.23%
Relative Difference	**0.00%**	**-19.28%**	**2.62%**	**0.10%**	**-0.18%**	**1.62%**	**1.26%**

Most relative differences due to variation in variogram parameters are less than 4%; well below the assessment CV (~30%)
Exception: search radius

Numbers in blue are those changing in the sensitivity

Variance

Kriging variance

Depends only on the distances between the data samples, not directly on the actual values

Is the variance of smoothed sample data and is different from the conventional sample variance

Other variance investigated

Jolly-Hampton on KS stratified results, uses INPFC regions as strata

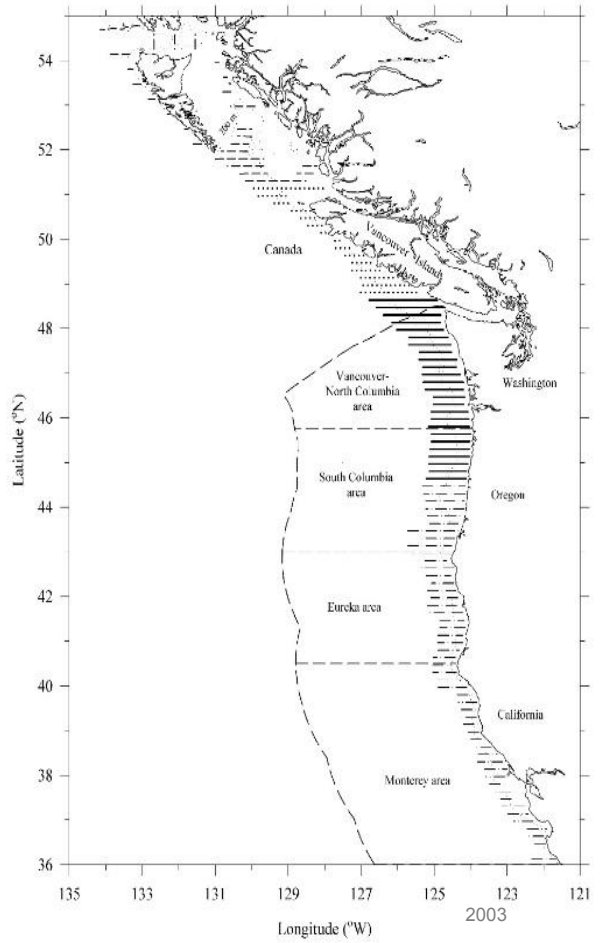



Photo by Ken High

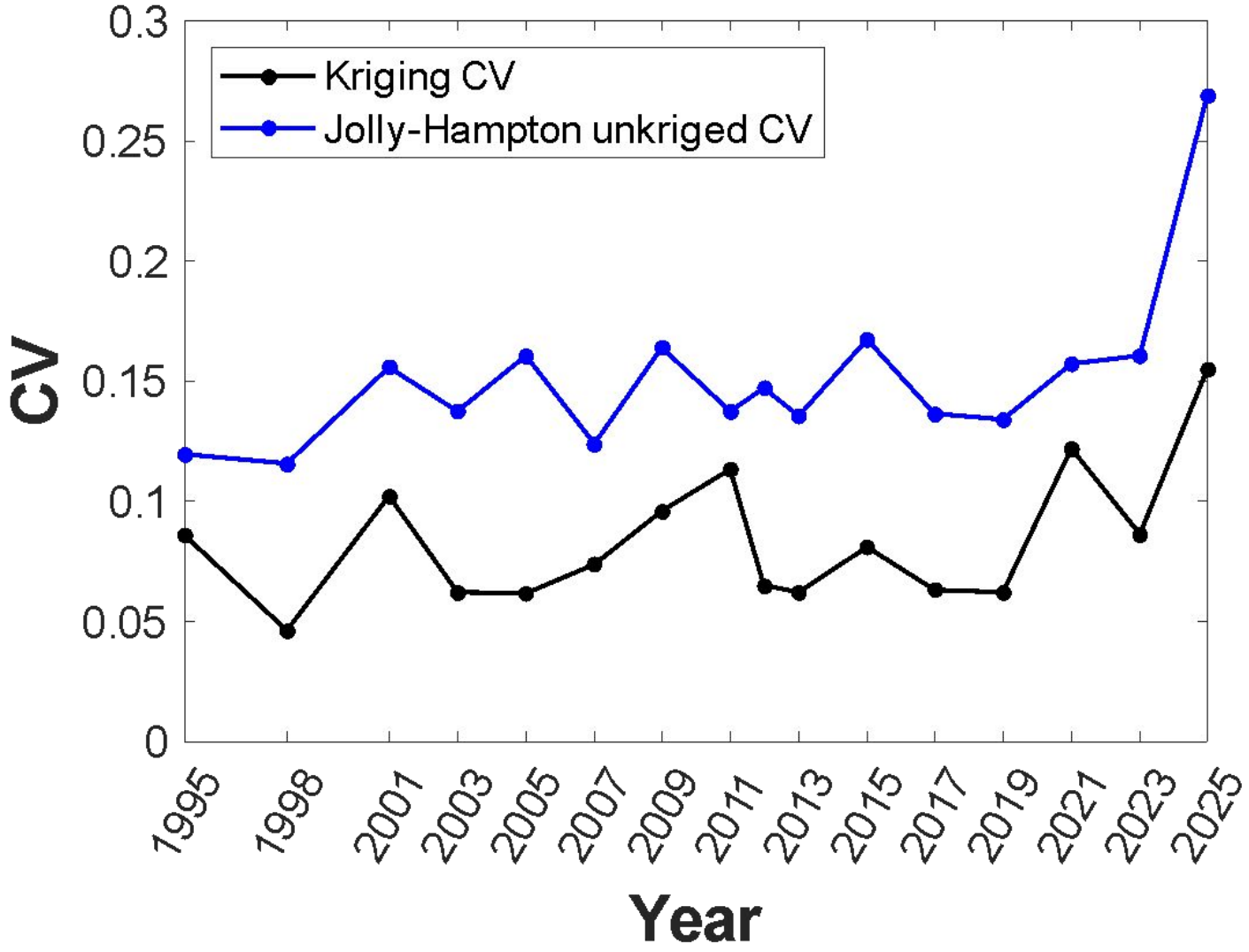
CV analysis – 2025 (backconverted to EK60)

Kriged	CV		Unkriged	CV
Biomass (mmt)	EchoPro	Jolly Hampton	Biomass (mmt)	Jolly Hampton
0.652	*15.49%*	*17.49**	*0.620*	*27.93%*

- EchoPro gives a kriging CV estimate
- Jolly-Hampton (JH) is a transect-based sample variance, normally larger than the kriging variance.
  - For the kriging JH estimate, 5 “virtual transects” per line of latitude were formed



Uncertainty estimates:



With vs without extrapolation

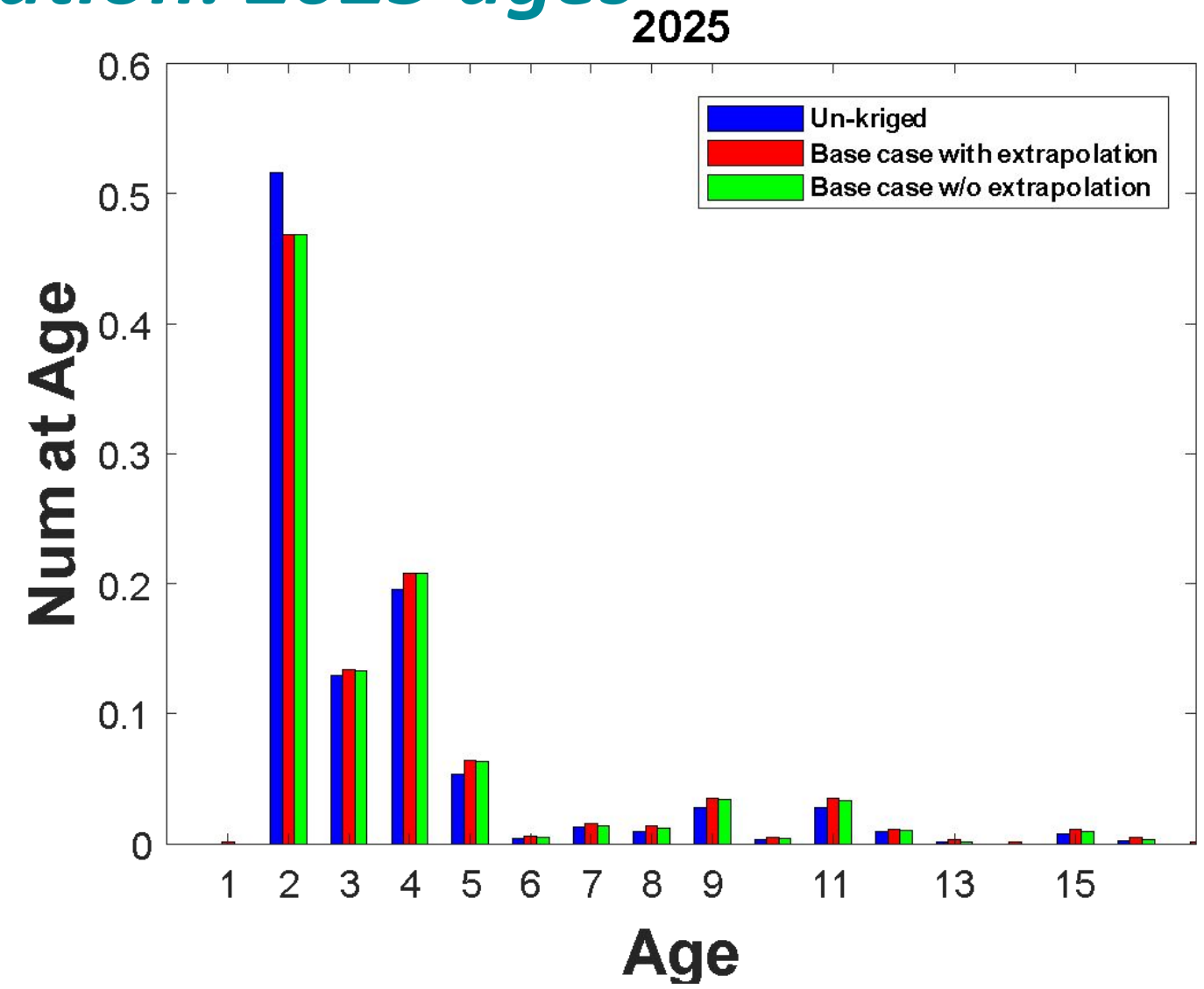
Stratification	Extrapolation	Un-kriged Biomass (mmt)	Kriged Biomass (mmt)	Relative Difference	CV (ALL)
KS	w/o	0.620	0.640	-1.76%	8.31%
	with		**0.652**	0.00%	15.49%

- **Un-kriged:** Linear interpolation of the area between transects and intervals along transects
- **Kriged without extrapolation:** Geostatistics-based interpolation of the area between transects and intervals along transects
- **Kriged with extrapolation:** same as above but with extrapolation beyond survey region
  - Base case: limited extrapolation using new kriging parameters

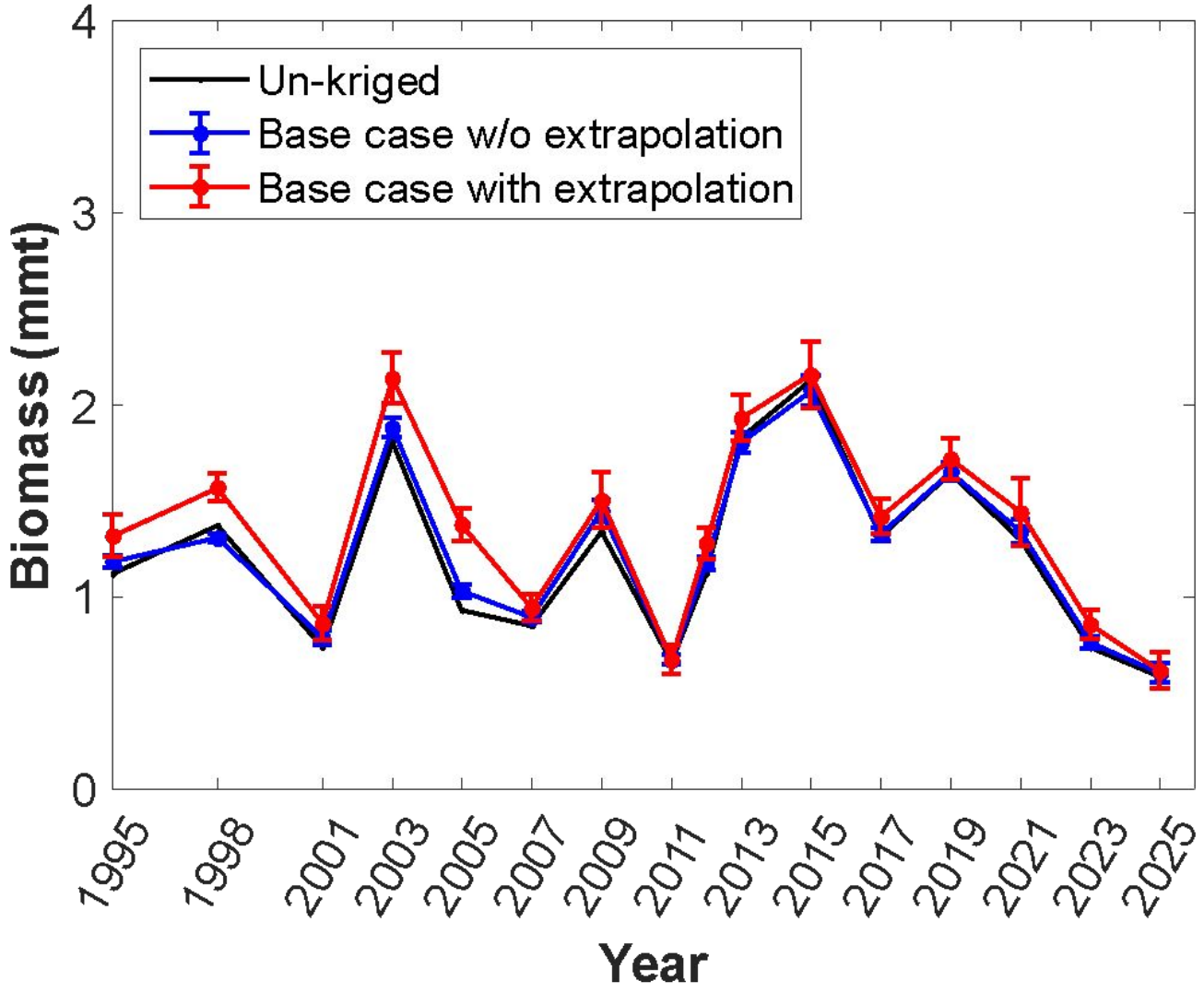
Biomass estimate time series

Year	Un-kriged (mmt)	Kriged (mmt) (Non-extrapolated)	**Kriged (mmt) (Base case)**	Kriging CV (US & CAN)
1995	1.119	1.184	**1.318**	8.59%
1998	1.371	1.305	**1.569**	4.60%
2001	0.738	0.787	**0.862**	10.20%
2003	1.807	1.880	**2.138**	6.19%
2005	0.931	1.030	**1.376**	6.16%
2007	0.853	0.894	**0.943**	7.38%
2009	1.338	1.448	**1.502**	9.57%
2011	0.662	0.671	**0.675**	11.33%
2012	1.124	1.174	**1.279**	6.47%
2013	1.830	1.803	**1.929**	6.20%
2015	2.128	2.072	**2.156**	8.09%
2017	1.318	1.325	**1.418**	6.32%
2019	1.657	1.643	**1.718**	6.20%
2021	1.375	1.423	**1.525**	12.18%
2023	0.780	0.809	**0.907**	8.61%
2025	0.620	0.640	**0.652**	15.49%

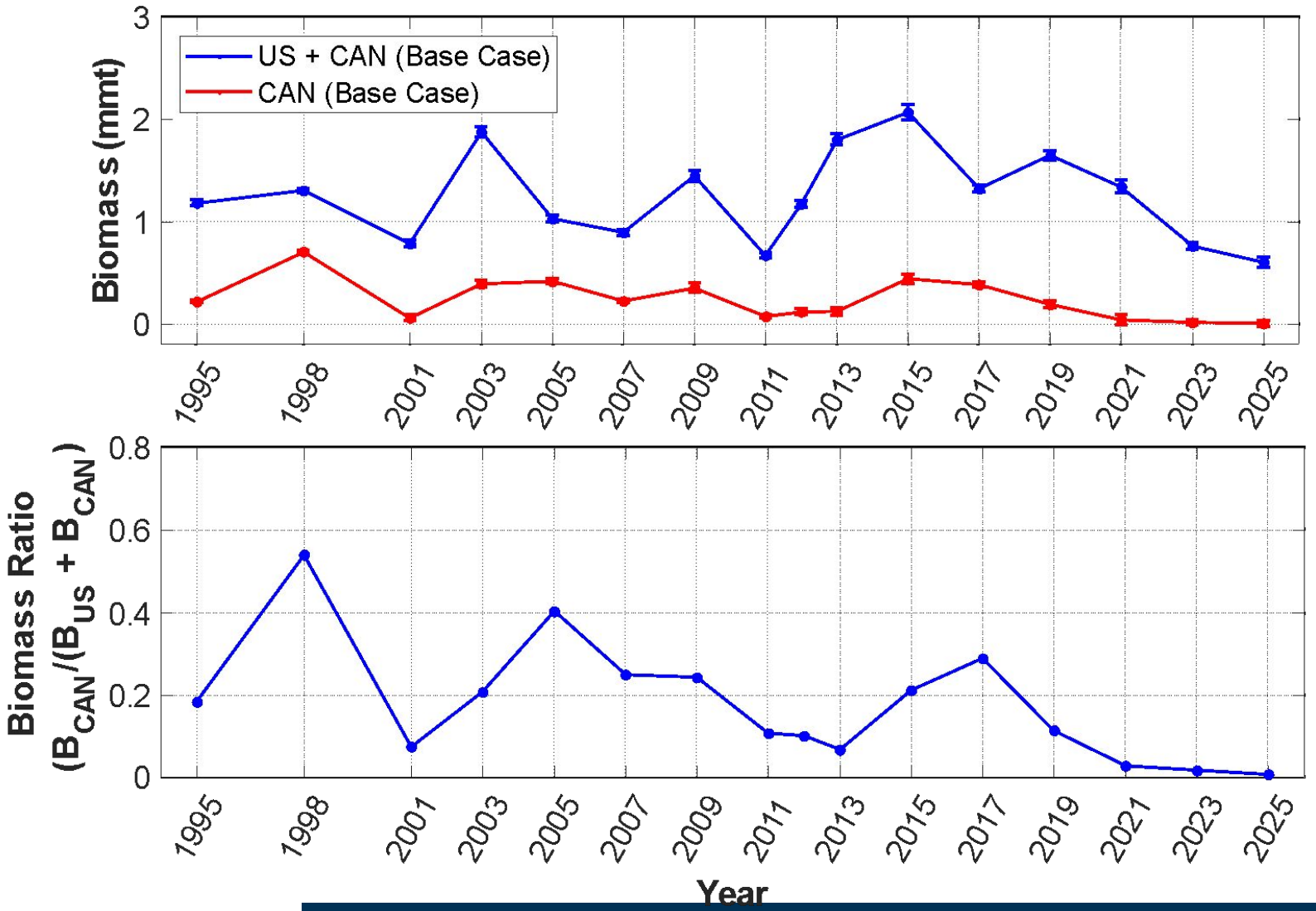
Extrapolation: 2025 ages



Extrapolation: survey time series



US and CAN estimates:



Summary: Biomass estimate/kriging


Photo by Rebecca Thomas

2025 Biomass estimate

0.652 mmt

Sensitivity analyses

Sensitivity to: aspect ratio & search radius

CV analysis

Kriging CV ~ 15.5%, unkriged Jolly Hampton ~ 28%

Non-extrapolation vs extrapolation

Consistent parameterizations to previous years

Extrapolated estimate slightly larger than non-extrapolated

EchoPop biomass estimation software

- Documentation: complete
  - https://echopop.readthedocs.io/en/latest/
- Time Series Validation: *in progress*
  - UW able to replicate 2015-2023 workflow, results match well
- Biological database pipeline: *early summer 2026*
  - Will improve efficiency of data ingestion and streamline workflow
- Selectivity: *summer 2026 (software capabilities)*
  - Implementation…
- FEAT validation and training: *in progress, continuing*